Section B will also make use of the resource booklet and candidates' own ideas and consist of a choice of *Going Global* or *World at Risk* longer/guided essay questions.

There are a total of 90 marks for the complete examination.

1.3 Topic 1: World at Risk

Global hazards	Global natural hazards vary in type and distribution and fall into two main categories — hydrometeorological and geophysical. The risks involved can turn hazards into natural disasters. This is especially true where a number of hazards occur together and where the population is already vulnerable – often as a result of high population density or poverty. These multiple hazard hotspots show how — when faced with disaster — the poor lose lives and the rich may lose money.
Climate change	Climate change is considered by many to be the world's greatest problem (technically a context hazard); and so a chronic, large-scale threat to people. Uncertainty about its impact is inevitable as scientists struggle to make firm predictions. It is an unfair world in which the wealthiest countries have emitted most of the greenhouse gases and the poorest ones are most vulnerable to their impacts. This topic poses questions about the causes of global warming, its relationship to long-term climate change, and the direct and indirect impacts that result. It also considers some of the solutions, whether global or local, and the complexity of managing international concerns against a background of national and personal self-interest. It also provides suggestions for tackling a variety of global hazards.

1 Global hazards

Enquiry question: *What are the main types of physical risks facing the world and how big a threat are they?*

What students need to learn	Suggested teaching and learning
■ Disasters result when hydro-meteorological hazards (cyclones, droughts and floods); and geophysical hazards (earthquakes, volcanoes and landslides/avalanches) threaten the life and property of increasing numbers of the world's people.	■ Exploring the concepts, processes and terminology relating to natural hazards, disasters and global warming.
■ The **R**isk of disaster grows as global **H**azards and people's **V**ulnerability increases, while their **C**apacity to cope decreases.	■ Making use of the disaster risk equation: $\mathbf{R = \dfrac{H \times V}{C}}$
■ Global warming arguably the greatest hazard we currently face is a chronic hazard; has widespread impacts; raises issues of injustice (polluters and vulnerable victims); and has complex solutions.	■ Assessing the status of global warming as the world's number one problem. (This is foundation work that may be used as a starting point or incorporated into the teaching of this unit as required, depending upon the prior learning of students.)

2 Global hazard trends

Enquiry question: *How and why are natural hazards now becoming seen as an increasing global threat?*

What students need to learn	Suggested teaching and learning
■ Some types of hazards are increasing in magnitude and frequency, and having greater impacts upon people and their lives.	■ Researching databases (eg CRED) for evidence of the size and frequency of the top six global natural hazards (cyclones, droughts, floods, earthquakes, volcanoes and landslides/ avalanches) upon lives, property, infrastructure and GDP.
■ Natural disasters are increasing because of a combination of physical and human factors ◆ the unpredictability of global warming and El Niño events leading to increasing natural hazards ◆ the increasing exploitation of resources (eg deforestation), world poverty, rapid population growth and urbanisation.	■ Exploring examples of how natural and human activities are combining to cause increasing disaster scenarios eg storms, floods and population change.
■ Trends show that the number of people killed is falling, whereas the number affected, and the economic losses are escalating.	■ Developing an awareness of how and why disasters are affecting more people and causing more damage yet lives are being saved, using examples of hazard events.

3 Global hazard patterns

Enquiry question: *Why are some places more hazardous and disaster-prone than others?*

What students need to learn	Suggested teaching and learning
■ An assessment of the real or potential natural hazard risks by using evidence about past or likely future events and their impact on people, property and the environment in their local area.	■ Exploring the local area to assess the risks from natural hazards such as flooding and global warming.
■ The distribution of the world's major natural hazards both hydrometeorological hazards and geophysical hazards (see 1).	■ Comparing and explaining various global distributions via maps and reports like the World Bank Hazard Management Unit.
■ Disaster hotspots occur when two or more hazards occur in vulnerable places: ◆ case study of the California coast ◆ case study of the Philippines; a vulnerable location.	■ Researching the causes, impacts and interaction of multiple hazards in contrasting hotspots.

4 Climate change and its causes

Enquiry question: *Is global warming a recent short term phenomenon or should it be seen as part of longer-term climate change?*

What students need to learn	Suggested teaching and learning
■ The current phenomenon of global warming should be set in the context of longer, medium and short term climate change. A range of evidence from ecology, historical records and climate change should be reviewed.	■ Researching the evidence of: ◆ longer-term, eg pollen analysis, ice cores, and past glacial/sea level change ◆ medium-term, eg historical records, tree rings and retreating glaciers ◆ recent, eg scientific research from weather, ocean, polar ice and ecosystem changes.
■ The causes of climate change may be both natural and human (anthropogenic).	■ Exploring the role of variations in earth orbit, solar output, cosmic collision and volcanic emissions, as well as enhanced greenhouse gas emissions.
■ Recent climate change (global warming) is unprecedented in historical terms and scientists now argue that human causes may be more to blame.	■ Assessing whether global warming is something unique or just a medium-term trend in the longer term pattern of climate variations.

5 The impacts of global warming

Enquiry question: *What are the impacts of climate change and why should we be concerned?*

What students need to learn	Suggested teaching and learning
■ The direct impacts of projected global climate changes: ◆ a case study of environmental and ecological impacts of Arctic warming in the Arctic region ◆ a case study of the complexities of economic impacts across the African continent and how it could lead to disasters for poor people.	■ Developing an awareness of the direct impacts of global warming through case studies of vulnerable places to understand the resulting environmental, ecological and economic impacts.
■ The indirect impacts such as the eustatic rise in sea level (global inundation).	■ Investigating how sea level rise may have a disproportionately bigger effect on some countries using examples such as the South Sea islands or Bangladesh (the disaster scenario).
■ The impacts of climate change are difficult to predict and emissions scenarios, such as the IPCC model, may vary (from 'business as usual' to sustainable) and could be affected by attempts to manage the impacts of climate change.	■ Researching the value of different scenarios and models in predicting future trends such as rising sea levels.
■ The evidence that combined impacts could lead to catastrophic, irreversible changes and contribute to a more hazardous world.	■ Exploring the concept of a 'tipping point'.

6 Coping with climate change

Enquiry question: *What are the strategies for dealing with climate change?*

What students need to learn	Suggested teaching and learning
■ How strategies: ◆ attempt to limit the impacts of climate change at various scales ◆ involve adapting to climate change.	■ Weighing up mitigation strategies and adaptation strategies using a range of examples of each.
■ The conflicting views and role of the key players in managing climate change — including governments, business, NGOs, individuals and groups. The complexities of a global agreement.	■ Developing an awareness of: ◆ the complexities of a global agreement such as the Kyoto protocol and its implications for specific countries ◆ national and small-scale strategies for limiting climate change ◆ the contributions of individuals to help reduce the impacts of climate change, eg carbon footprints.
■ Whilst most people argue for 'act local, think global', management is needed at all scales and progress is likely to be incremental.	

7 The challenge of global hazards for the future

Enquiry question: *How should we tackle the global challenges of increasing risk and vulnerability in a more hazardous world?*

What students need to learn	Suggested teaching and learning
■ Increasing risk and uncertainty threatens major disruption to people and the environment at a global scale bringing water shortages and food insecurity.	■ Developing an awareness of how other global problems (eg conflict, famine, climate change and poverty) make managing global hazards more difficult.
■ The world should recognise that global warming is one of the biggest challenges it has faced and make innovative choices, adopt sustainable strategies and understand the cost and benefits involved.	■ Investigating and weighing up strategies to manage global warming, such as energy efficiency, conservation, decreasing carbon emissions, alternative energy and reafforestation.
■ Solutions to a hazardous world, at all scales, need to focus on the underlying issues of risk and vulnerability.	■ Developing an awareness of, for example, local flood risk, regional poverty and international strategies to tackle a world at risk.